But $OQ = \text{radius} = 1$

$\therefore \quad OP = OQ \cos\theta = \cos\theta$

Similarly, $\dfrac{PQ}{OQ} = \sin\theta$

$\Rightarrow PQ = OQ \sin\theta = \sin\theta \ (\because OQ = 1)$

$OP = \cos\theta,\ PQ = \sin\theta \quad \ldots (1)$

Now, from triangle QOR,

we have $\dfrac{OR}{OQ} = \cos(90° - \theta)$

$\therefore \ OR = OQ \cos(90° - \theta)$

$OR = \cos(90° - \theta)$

Similarly, $\dfrac{RQ}{OQ} = \sin(90° - \theta)$

Then, $RQ = \sin(90° - \theta)$

$OR = \cos(90° - \theta),\ RQ = \sin(90° - \theta) \quad \ldots (2)$

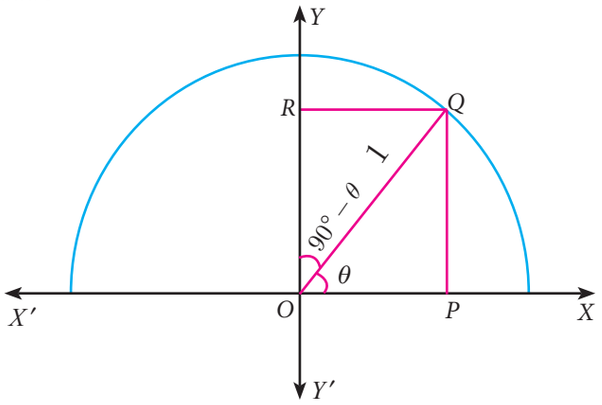


Fig. 6.2

$\because OPQR$ is a rectangle,

$OP = RQ$ and $OR = PQ$

Therefore, from (1) and (2) we get,

$\sin(90° - \theta) = \cos\theta$ and $\cos(90° - \theta) = \sin\theta$

Note

$(\sin\theta)^2 = \sin^2\theta$	$(\text{cosec}\theta)^2 = \text{cosec}^2\theta$
$(\cos\theta)^2 = \cos^2\theta$	$(\sec\theta)^2 = \sec^2\theta$
$(\tan\theta)^2 = \tan^2\theta$	$(\cot\theta)^2 = \cot^2\theta$

Table of Trigonometric Ratios for $0°, 30°, 45°, 60°, 90°$

θ / Trigonometric Ratio	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined
$\text{cosec}\theta$	undefined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	undefined
$\cot\theta$	undefined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

6 Trigonometry.indd 241 15-12-2020 03:13:26 PM